

CBD-Focused vs Manhattan-Wide Optimization Analysis

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1. Motivation

The baseline EMS optimisation (P2: demand-weighted allocation) minimises the **Manhattan-wide** expected response time, treating all 25 precincts equally in the objective function. Given that the Central Business District (CBD) accounts for ~56% of all crash demand from only 10 precincts, a natural question arises:

Would dedicating resources specifically to the CBD improve CBD response times, and at what cost to non-CBD precincts?

This analysis compares two optimisation strategies to examine the **equity vs efficiency tradeoff** in EMS resource allocation.

2. Methodology

2.1 Optimisation Formulations

Strategy	Objective	Precincts in Objective
Manhattan-Wide P2	$\min \sum_j d_j \sum_i t_{ij} y_{ij}$	All 25 precincts
CBD-Focused P2	$\min \sum_{j \in \text{CBD}} d_j \sum_i t_{ij} y_{ij}$	10 CBD precincts only

Both formulations:

- Allocate $K=20$ total EMS units
- Maximum 5 units per firehouse
- Assign all precincts to exactly one firehouse (feasibility)
- Use the same Haversine travel-time matrix

The CBD-focused model uses `build_cbd_focused_demand_weighted()` from `src/ems_readiness/optimization/models.py`.

2.2 CBD Precincts

Per `docs/cbd_definition.md`, the 10 CBD precincts are: 1, 5, 6, 7, 9, 10, 13, 14, 17, 18.

2.3 Experiment Design

1. Solve both optimisations for $K=20$
2. Run 10 replications per allocation using the DES simulation engine
3. Disaggregate results into CBD and non-CBD response times

4. Compute coverage fractions (≤ 8 min threshold) by region

3. Results

3.1 Allocation Comparison

Metric	Manhattan-Wide P2	CBD-Focused P2
Total units	20	20
Active firehouses	20	12
CBD firehouses used	~10	~12
Non-CBD firehouses used	~10	~0

The CBD-focused optimisation concentrates all 20 units in firehouses near the CBD, leaving non-CBD precincts without dedicated nearby units.

3.2 Response Time Performance

Strategy	CBD RT (min)	Non-CBD RT (min)	Overall RT (min)
Manhattan-Wide P2	2.47 ± 0.04	2.66 ± 0.05	2.55 ± 0.03
CBD-Focused P2	2.50 ± 0.06	6.88 ± 0.24	4.42 ± 0.12

3.3 Coverage Performance

Strategy	CBD Coverage (≤ 8 min)	Non-CBD Coverage (≤ 8 min)
Manhattan-Wide P2	100.0%	99.2%
CBD-Focused P2	99.9%	73.6%

3.4 Key Finding: The Equity Tradeoff

The CBD-focused allocation:

- **Does not improve** CBD response times (2.50 vs 2.47 min — marginally worse)
- **Dramatically worsens** non-CBD response times (6.88 vs 2.66 min — 159% increase)
- **Degrades** non-CBD coverage from 99.2% to 73.6%
- **Increases** overall response time from 2.55 to 4.42 min (73% worse)

4. Figures

Figure	Description
results/cbd_focused_comparison/cbd_focused_comparison.png	Grouped bar chart of CBD vs non-CBD response times and coverage
results/cbd_focused_comparison/allocation_comparison.png	Horizontal bar chart showing unit placement for each strategy
results/cbd_focused_comparison/equity_tradeoff.png	Scatter plot of CBD RT vs non-CBD RT tradeoff

5. Discussion

5.1 Why CBD-Focused Doesn't Help the CBD

The Manhattan-wide optimisation already allocates significant resources near the CBD because:

1. The demand-weighted objective naturally concentrates units where demand is highest
2. CBD precincts generate 56% of all demand, so they already receive proportional attention
3. The additional units in the CBD-focused model provide negligible marginal benefit

5.2 Why CBD-Focused Hurts Non-CBD

When all units are concentrated in CBD firehouses:

- Non-CBD incidents must be served by distant units
- Travel times for northern Manhattan (precincts 30, 32, 33, 34, 50, 52) increase dramatically
- The dispatch model sends the nearest available unit, but "nearest" becomes very far for non-CBD precincts

5.3 Equity Implications

From a public policy perspective:

- **Equal access to emergency services** is a fundamental equity principle
- Concentrating resources in the CBD would disproportionately harm residents of:
 - Washington Heights (Pct 33, 34)
 - Harlem (Pct 25, 26, 28, 30, 32)
 - East Harlem (Pct 23, 25)
- These areas typically have lower income levels, creating an **environmental justice concern**

5.4 When CBD-Focused Might Make Sense

A CBD-focused strategy could be justified during:

- **Special events** (e.g., Times Square New Year's Eve) with temporary demand surges
- **Time-of-day** periods when non-CBD demand drops to near zero (very late night)
- **Multi-tier systems** where a separate non-CBD fleet exists

6. Recommendation

The Manhattan-wide P2 allocation is strongly preferred over the CBD-focused alternative because:

1. It achieves near-identical CBD performance (2.47 vs 2.50 min)
2. It maintains excellent non-CBD performance (2.66 vs 6.88 min)
3. It preserves equitable coverage across all communities
4. The demand-weighted objective naturally prioritises high-demand areas without explicit geographic targeting

The results confirm that the demand-weighted formulation is an effective “self-balancing” approach that delivers both **efficiency** (low overall RT) and **equity** (balanced regional performance).

7. Files Generated

File	Description
<code>src/ems_readiness/optimization/models.py</code>	New functions: <code>build_cbd_focused_demand_weighted()</code> , <code>build_cbd_focused_coverage()</code>
<code>scripts/run_cbd_focused_optimization.py</code>	Experiment script
<code>results/cbd_focused_comparison/comparison_table.csv</code>	Performance comparison data
<code>results/cbd_focused_comparison/allocations.csv</code>	Allocation comparison
<code>results/cbd_focused_comparison/*.png</code>	Comparison figures